

```
1 package ques3SortingCustomerOrders;
2
3 public class Order {
4     int orderId;
5     String customerName;
6     double totalPrice;
7
8     public Order(int orderId, String customerName, double totalPrice) {
9         this.orderId = orderId;
10        this.customerName = customerName;
11        this.totalPrice = totalPrice;
12    }
13
14    @Override
15    public String toString() {
16        return "Order ID: " + orderId +
17            ", Customer: " + customerName +
18            ", Total Price: " + totalPrice;
19    }
20
21 }
22
```

```
1 package ques3SortingCustomerOrders;
2 public class SortOperations {
3
4     // Bubble Sort
5     public static void bubbleSort(Order[] orders) {
6
7         int n = orders.length;
8
9         for (int i = 0; i < n - 1; i++) {
10
11             for (int j = 0; j < n - i - 1; j++) {
12
13                 if (orders[j].totalPrice > orders[j + 1].totalPrice) {
14
15                     Order temp = orders[j];
16                     orders[j] = orders[j + 1];
17                     orders[j + 1] = temp;
18                 }
19             }
20         }
21     }
22
23     // Quick Sort
24     public static void quickSort(Order[] orders, int low, int high) {
25
26         if (low < high) {
27
28             int pivotIndex = partition(orders, low, high);
29
30             quickSort(orders, low, pivotIndex - 1);
31             quickSort(orders, pivotIndex + 1, high);
32         }
33     }
34 }
```

```

35 private static int partition(Order[] orders, int low, int high) {
36
37     double pivot = orders[high].totalPrice;
38
39     int i = low - 1;
40
41     for (int j = low; j < high; j++) {
42
43         if (orders[j].totalPrice < pivot) {
44
45             i++;
46
47             Order temp = orders[i];
48             orders[i] = orders[j];
49             orders[j] = temp;
50         }
51     }
52
53     Order temp = orders[i + 1];
54     orders[i + 1] = orders[high];
55     orders[high] = temp;
56
57     return i + 1;
58 }
59
60 // Display Orders
61 public static void display(Order[] orders) {
62
63     for (Order order : orders) {
64         System.out.println(order);
65     }
66 }
67 }

```

```
package ques3SortingCustomerOrders;
public class SortTest {

    public static void main(String[] args) {

        Order[] orders = {

            new Order(101, "Amit", 2500),
            new Order(102, "Rahul", 1200),
            new Order(103, "Priya", 5000),
            new Order(104, "Neha", 3000)
        };

        System.out.println("Before Sorting:");
        SortOperations.display(orders);

        SortOperations.bubbleSort(orders);

        System.out.println("\nAfter Bubble Sort:");
        SortOperations.display(orders);

        Order[] orders2 = {

            new Order(101, "Amit", 2500),
            new Order(102, "Rahul", 1200),
            new Order(103, "Priya", 5000),
            new Order(104, "Neha", 3000)
        };

        SortOperations.quickSort(orders2, 0, orders2.length - 1);

        System.out.println("\nAfter Quick Sort:");
        SortOperations.display(orders2);
    }
}
```



Before Sorting:

Order ID: 101, Customer: Amit, Total Price: 2500.0

Order ID: 102, Customer: Rahul, Total Price: 1200.0

Order ID: 103, Customer: Priya, Total Price: 5000.0

Order ID: 104, Customer: Neha, Total Price: 3000.0

After Bubble Sort:

Order ID: 102, Customer: Rahul, Total Price: 1200.0

Order ID: 101, Customer: Amit, Total Price: 2500.0

Order ID: 104, Customer: Neha, Total Price: 3000.0

Order ID: 103, Customer: Priya, Total Price: 5000.0

After Quick Sort:

Order ID: 102, Customer: Rahul, Total Price: 1200.0

Order ID: 101, Customer: Amit, Total Price: 2500.0

Order ID: 104, Customer: Neha, Total Price: 3000.0

Order ID: 103, Customer: Priya, Total Price: 5000.0